

KnowServe Corp. — Research & Development Department

Experimental Results Dataset — Q1 2025

Objective

Summarize outcomes of three parallel experiments conducted under Project ORBIT.

Dataset Statistics

- Documents Processed – 1,000,000
- Average Tokens per Document – 412
- Embedding Vector Dimensionality – 1,536
- Model Version – Orbit-BERT v2.1
- Hardware – 4× A100 GPUs, 64 GB RAM

Results

- Mean Retrieval Accuracy – 92.4 % (↑ 3.1 %)
- Average Response Latency – 176 ms (↓ 18 %)
- Energy Consumption – 0.81 kWh per 10k queries (↓ 11 %).

Observations

Experiment 3 achieved best trade-off between accuracy and latency using dynamic quantization.

Outlier analysis revealed bias toward English-dominant texts → language balancing planned for Q2.

Storage & Access

All datasets stored in AWS S3 bucket rd-orbit-datasets (region eu-central-1) with public-read disabled.

Checksum and metadata published on internal portal /rd/results/q1-2025.